Same Question, Different Prompts: How Prompt Variability Affects RAG Retrieval and Hallucination

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Abstract

Retrieval-augmented generation (RAG) uses documents or data as "knowledge" or "context" to help a language model answer questions. However, sometimes even when two questions mean the same thing, changing the wording of the question can lead to different retrieval results. In this project, I examined how this affects retrieval and answer quality using 28 questions from HotpotQA. I tested each question in its original base form, and with three rewritten or paraphrased versions, thus giving me 112 question cases in total. I used sentence embeddings and FAISS for retrieval. I used Qwen2.5-0.5B-Instruct to generate answers. I also tested mitigation methods, such as evidence-constrained generation, query rewriting, and cross-encoder reranking, to see if they could improve the results. The retrieval results varied across the different versions of the questions. All supporting titles were retrieved for 42.86% of the original questions, compared with 32.14% to 35.71% of the rewritten versions. However, the differences in answer correctness and evidence support were not statistically significant. Evidence-constrained generation gave the best results for fully correct and fully supported answers among the generation methods I tested. Query rewriting improved some aspects of answer correctness but produced fewer fully supported answers. Reranking improved retrieval, and Recall@5 from 66.52% to 77.68%. Overall, the experiments do show us that changing a question's wording can affect which passages the system retrieves. It also tells us that better retrieval does not necessarily lead to a better answer, so retrieval performance and answer quality both need to be tested separately.

1 Introduction

Retrieval Augmented Generation (RAG) is one of the techniques which is used to enhance the performance of language models. A RAG technique is not just relying on the prior training data to generate answers, but it first selects relevant documents that are later used to generate the answers. This attempts to ensure the generation of useful answers as well as reducing the risk of incorrect statements. However, the quality of the answer is heavily determined by the quality of retrieved evidence.

The first aspect which affects retrieving is the way in which a question is formulated. On a daily basis, people ask the same question differently, while they all might mean the same thing. Similarly, we can paraphrase a question and ask it in a different manner. Even though such versions of a question can appear to a person to be equivalent, an RAG system might not consider them in the same manner. A slight change in the wording can mean the difference in identifying relevant passages and providing a language model with appropriate or inappropriate information.

This leads to an important question about the reliability of RAG systems: if the wording of a question changes, but its meaning stays the same, will the system retrieve the same supporting information and produce a similarly reliable answer? If important evidence is missed during retrieval, the language model may give an incomplete answer or generate a response that is not properly supported by the available context (which we will later discuss as hallucinations). This makes prompt variability an important topic in terms of RAG systems and LLMs.

In this project, I have assessed how different formulations of the same question affect retrieval performance and the quality or correctness of generated answers. A curated set of questions from HotpotQA is our basis, with three rewritten versions of each original question. I will then compare the retrieval results across these versions to examine whether changes in wording actually lead to differences in the evidence retrieved by the system. Examining the generated answers to understand

how retrieval differences are related to answer correctness and evidential support is also important.

Aside from studying prompt variability, I will also evaluate a few approaches that may help improve RAG performance. These include evidence-constrained generation, query rewriting, and retrieval reranking. Each approach addresses a different part of the process, such as controlling how the model uses retrieved evidence, changing the query before retrieval, or improving the order of the retrieved passages. By comparing these approaches, I can attempt to understand which aspects of the RAG pipeline have the greatest influence on the quality of the final answers.

The main research questions addressed in this project are:

- How does changing the wording of a question affect the retrieval of supporting passages in a RAG system?
- Are changes in retrieval performance affecting the correctness and evidential support of the generated answers?
- How do evidence-constrained generation and query rewriting affect the quality and reliability of generated answers?
- Does retrieval reranking improve the system's ability to retrieve supporting evidence?

2 Background and Related Work

2.1 Retrieval-Augmented Generation

LLMs can generate answers, but sometimes they may produce incorrect or unsupported information. RAG addresses this by adding to a language model an external information retrieval system. Before generating an answer, the system retrieves relevant passages from a document collection and provides them to the model as context. This method was introduced by Lewis et al. (2020) and has become an important method for question answering that require knowledge or context of anything.

That being said, the quality of a RAG system depends heavily on the information it retrieves. If evidence is missing, the language model may generate an incomplete answer or make unsupported claims, which are both important issues.

2.2 Dense Retrieval and Prompt Variability

Traditional retrieval methods match words between a query and a document. Dense retrieval methods, on the other hand, convert queries and passages into numerical vectors. Thus, they are matched based on semantic similarity(Karpukhin et al., 2020).

Even though dense retrieval identifies passages with similar meanings, if a question is changed in terms of its wording, it may affect the retrieved results. This is important because the same question can be expressed in several ways without changing its intended meaning. Papers such as PromptBench by **?** also highlight the importance of checking these.

2.3 Hallucination in RAG Systems

Hallucination is a specific case where a language model produces an answer that is incorrect or not supported by the available evidence(Ji et al., 2023). This is a concern in RAG systems as well, since retrieved passages do not guarantee that the model will stick to them(Niu et al., 2024).

2.4 Improving RAG Performance

Different methods can be used to improve the retrieval and generation of an RAG system. Query rewriting tries to improve the wording of a question before retrieval. Reranking changes the order of retrieved passages to put more relevant evidence higher. Evidence-constrained generation encourages the language model to answer using only the supplied context.

3 Dataset and Data Preparation

3.1 Dataset Overview

For this project, I used a sample of the HotpotQA dataset. HotpotQA is a question-answering dataset where many questions require information from more than one passage to arrive at an answer(Yang et al., 2018). This made it suitable for my experiments, since I wanted to see how changes in question wording affect the retrieval of supporting information.

The source sample contained 1,000 question-answer examples. Along with these examples, I prepared a passage collection using the titles and sentences available in their contexts. After removing duplicate titles, the collection contained 9,758 unique passages. This became the document collection searched by the RAG system.

3.2 Selection of Questions

I did not use all 1,000 questions for the main experiments. Instead, I initially selected 30 questions

for a closer evaluation. The selection included two types of questions: bridge questions, which require connecting information from different sources, and comparison questions, which involve comparing two or more entities.

During the review of these questions, I found two cases where the available passages were not enough for a reliable evaluation. One question contained an ownership premise that did not match with the evidence. In another case, the passages did not exactly state the claimed equality between two canal lengths. I removed both questions rather than including cases where the expected answer could not be properly checked against the available information.

This left me with 28 questions for the final evaluation. Of these, 17 were bridge questions and 11 were comparison questions. The difficulty distribution was 2 easy, 18 medium, and 8 hard questions.

3.3 Creating Different Versions of Each Question

The main part of my experiment was to compare different ways of asking the same question. For every question in the final set, I kept the original wording and prepared three rewritten versions. The intention was to change how the question was expressed while keeping the information being asked for as close as possible to the original.

The three prompts created were Paraphrased (slightly different wording), Extended (much longer prompt), and Distracted (irrelevant information in with prompt).

This resulted in four versions of each question. Four prompts per question, and 28 questions, so there were 112 prompt cases in total.

3.4 Summary of the Evaluation Set

Table 1 is an overview of the data used in the final experiments.

Dataset characteristic	Count
Source question sample	1,000
Unique passages	9,758
Initially selected questions	30
Excluded questions	2
Final questions	28
Bridge questions	17
Comparison questions	11
Easy questions	2
Medium questions	18
Hard questions	8
Versions per question	4
Total prompt cases	112

Table 1: Overview of the dataset and final evaluation set.

4 Methodology and Experimental Setup

4.1 Overall System Setup

I built the RAG system around a relatively simple pipeline. First, A question is converted into an embedding. Then that embedding is used to search the passage collection. The most relevant passages are given to the language model, which uses them to generate an answer.

For the retrieval part, I used the `all-MiniLM-L6-v2` model from Sentence Transformers. Each passage was represented using its title and text, and the resulting embeddings were normalised before being added to a FAISS index. I used FAISS with an `IndexFlatIP` index, which compares the normalised vectors using inner product. Since the vectors are normalised, this is equivalent to cosine similarity.

The system retrieved the top five passages for each question. These passages were then used as context for answer generation. The same retrieval setup was used across the four prompt versions in the main experiment so that I could compare the effect of wording without intentionally changing the retrieval configuration.

4.2 Answer Generation

For answer generation, I used the `Qwen2.5-0.5B-Instruct` model through the Hugging Face Transformers library. The model was loaded in half-precision (FP16) and run on my laptop's RTX 3050Ti.

I kept the generation settings fixed across the experiments. The model could generate up to 48 new tokens, and greedy decoding was used instead of sampling. A repetition penalty of 1.05 was also applied. The input length was limited to 8,192 tokens, while each retrieved passage was limited to 1,500 characters before being added to the context.

The baseline instruction asked the model to answer the question using the supplied documents and to keep the response short and direct. It was also told not to deliberately guess when the documents did not contain enough information. I kept the answers relatively short because the main aim was to evaluate whether the model could identify and use the required information.

4.3 Prompt Variability Experiment

The first experiment focused on whether changing the wording of a question affected retrieval. I used all 28 questions, with the original version and three rewritten versions of each question. This gave me 112 prompt cases.

Each version was passed through the same embedding and retrieval process. I then compared the retrieved passage titles with the supporting titles provided for the question. This allowed me to check whether the required evidence appeared among the top five results.

I used three main retrieval measures. Recall@5 measured the proportion of supporting titles found in the first five results. Complete retrieval measured whether all the required supporting titles were found within those results. MRR@5 measured the rank of the first supporting passage, giving a higher score when relevant evidence appeared closer to the top.

I also calculated the overlap between the top five results for each rewritten prompt and those retrieved for the original question. This gave me another way to observe how much the retrieved results changed after the question was reworded.

4.4 Answer Quality Evaluation

Once I was done with retrieval, I examined the answers generated for the different prompt versions. I did not rely on correctness alone, since an answer can actually be factually correct without being supported by the passages given to the model.

I therefore evaluated correctness and evidence support separately. Correctness measured whether the generated answer matched the expected answer. Evidence support focused on whether the retrieved context actually provided support for the claims made in the response.

I also recorded the main type of failure in cases where the answer was not satisfactory. The categories included missing information during retrieval, errors during answer generation, and incomplete answers.

Hallucination was examined separately through an annotation process. I checked the generated responses for information that was unsupported by the available evidence. This distinction is important because an incorrect answer and an unsupported answer are not the same thing.

4.5 Comparison of Generation Methods

Next, I compared the baseline system with two alternative methods: evidence-constrained generation and query rewriting.

For evidence-constrained generation, I kept the same Qwen model but changed the instructions given to it. The model was explicitly told to use only the supplied documents, avoid outside knowledge and assumptions, as well as to state when the evidence was insufficient. It was also instructed not to guess. Using this I could check whether clearer restrictions could make the generated answers more reliable.

For query rewriting, I used Qwen again, this time to produce a rewritten search query before retrieval. The rewriting instructions asked the model to preserve the original question's intent and all other important data, like names or dates. The rewritten query was then embedded using `all-MiniLM-L6-v2`, and FAISS was used to retrieve the top five passages. The final answer was generated for the original question using the retrieved context.

I compared these methods using the number of fully correct answers, the number of answers that were at least partly correct, and the number of fully supported answers. I also compared their hallucination annotations to see whether the different instructions had an effect on unsupported content.

4.6 Retrieval Reranking

The final experiment focused on improving the order of retrieved passages. I used a cross-encoder model, `cross-encoder/ms-marco-MiniLM-L-6-v2`, as a reranker.

The system first retrieved 20 candidate passages using dense retrieval. The cross-encoder then scored each query and passage pair. The cross-encoder considers the query and passage together when assigning a relevance score. The 20 candidates were sorted again using these scores, and the top five were selected as the final results.

I compared retrieval performance before and after reranking using Recall@5, complete retrieval,

and MRR@5. I also examined the answers generated from the reranked passages to see whether the improved ordering of evidence led to better answer quality.

4.7 Statistical Analysis

Since the same 28 questions were used across the different prompt versions and methods, I used paired comparisons rather than treating the results as independent groups. For binary outcomes, such as whether all supporting titles were retrieved or whether an answer was fully correct, I used the exact McNemar test.

The resulting p-values helped me judge whether the differences between approaches were statistically significant. I treated the reranking retrieval results mainly as a comparison, since it was a mitigation or improvement method for the RAG system.

5 Experiments and Results

This section presents the results from the prompt variability experiment and the three methods tested to improve retrieval or answer quality. I am going to report retrieval performance and answer quality separately, because retrieving a supporting passage does not always mean that the model will use it correctly. The experiments use the final set of 28 questions and their four prompt versions, giving 112 prompt cases in total.

5.1 Effect of Prompt Wording on Retrieval

In the first experiment, the main point to check was if changing the wording of a question affected the passages retrieved by the system. I compared the original question with its three altered versions using Recall@5, complete retrieval of supporting titles, Mean Reciprocal Rank (MRR), and top-five overlap with the original prompt.

The original questions achieved a mean Recall@5 of 69.64%. The three rewritten versions scored 64.29%, 66.07%, and 66.07%, respectively. The original wording also retrieved all supporting titles for 12 of the 28 questions. This fell to 9, 10, and 10 questions for the three rewritten versions.

Prompt	R@5	All titles	MRR	Overlap
Original	69.64%	12/28	0.8929	100%
P1	64.29%	9/28	0.8839	86.43%
P2	66.07%	10/28	0.8512	82.86%
P3	66.07%	10/28	0.8601	77.86%

Table 2: Retrieval performance across the four prompt versions. R@5 is Recall@5; all titles means the complete retrieval of supporting titles; overlap is measured with the original prompt's top five results.

The overlap scores show that the retrieved results were not identical across prompt versions. Paraphrase 3 had the lowest overlap with the original results, at 77.86%. This suggests that even when the question meaning was intended to remain the same, its wording could change which passages appeared in the top five.

However, the differences in complete supporting-title retrieval were not statistically significant in the paired exact McNemar tests. The p-values for the three rewritten versions compared with the original were 0.250, 0.500, and 0.727. Therefore, the results show variation in retrieval, but they do not provide strong statistical evidence that the rewritten prompts performed worse on this sample.

5.2 Answer Quality Across Prompt Versions

I also compared the answers generated from each prompt version. I evaluated correctness separately from evidential support. An answer could contain the expected answer but still lack enough support from the retrieved passages, so treating these as the same measure would hide an important part of the system's behaviour.

The original prompts produced 8 fully correct answers out of 28 (28.57%). Paraphrase 1 also produced 7 fully correct answers (25.00%), while Paraphrase 2 produced 8 (28.57%). Paraphrase 3 produced 6 (21.43%).

The pattern was similar for fully supported answers, although the counts were lower. The original prompts produced 7 fully supported answers (25.00%). The three rewritten versions produced 5 (17.86%), 6 (21.43%), and 6 (21.43%), respectively.

Prompt	**Fully correct**	**Fully supported**
Original	8/28 (28.57%)	7/28 (25.00%)
Paraphrase 1	7/28 (25.00%)	5/28 (17.86%)
Paraphrase 2	8/28 (28.57%)	6/28 (21.43%)
Paraphrase 3	6/28 (21.43%)	6/28 (21.43%)

Table 3: Answer correctness and evidential support across prompt versions.

The paired tests did not show statistically significant differences in fully correct answers between the original prompts and any of the rewritten versions. The p-values were 1.000, 1.000, and 0.625. The support results also did not show a significant prompt-level difference in the paired comparisons.

These results do not mean that prompt wording had no effect. The retrieval results show that the system returned different passages for some rewritten questions. However, those retrieval changes did not translate into a consistent or statistically significant difference in answer correctness in this experiment. The relatively low number of fully correct answers also suggests that the task was not very easy for the small model used.

Failure category	Cases	Percentage
Generation errors	43	38.39%
Retrieval misses	42	37.50%
Incomplete answers	7	6.25%
No failure label	20	17.86%
Total	112	100.00%

Table 4: Distribution of the main failure categories in the answer audit. Each case was assigned a dominant category.

5.3 Comparison of Answer Generation Methods

The next experiment compared the baseline system with evidence-constrained generation and query rewriting. Each method was evaluated on the same 112 prompt cases. I used three metrics: fully correct answers, answers that were at least partly correct, and fully supported answers.

Evidence-constrained generation achieved the highest number of fully correct answers, with 44 out of 112 (39.29%). Query rewriting gave about 42 (37.50%). The baseline was 29 (25.89%). Evidence-constrained generation also had the highest number of answers that were at least partly correct.

Method	Correct	Partial+	Supported
Baseline	29/112	48/112	24/112
Evidence-constr.	44/112	55/112	37/112
Query rewrite	42/112	47/112	12/112

Table 5: Answer quality across generation methods. Values are counts out of 112. Correct means fully correct answers, Partial+ means at least partly correct answers, and Supported means fully supported answers. The corresponding percentages are 25.89%, 42.86%, and 21.43% for the baseline; 39.29%, 49.11%, and 33.04% for evidence-constrained generation; and 37.50%, 41.96%, and 10.71% for query rewriting.

The paired tests found a significant improvement in fully correct answers for evidence-constrained generation over the baseline ($p = 0.0041$). The baseline comparison to query-rewriting was also significant ($p = 0.0470$). There was no significant difference in fully correct answers between evidence-constrained generation and query rewriting ($p = 0.8388$).

The results for fully supported answers were different. Evidence-constrained generation achieved 37 fully supported answers, compared with 24 for the baseline and 12 for query rewriting. The paired comparisons showed significant differences between all three method pairs: baseline versus evidence-constrained generation ($p = 0.0106$), baseline versus query rewriting ($p = 0.0290$), and evidence-constrained generation versus query rewriting ($p = 0.000041$).

This distinction is important. Query rewriting improved the number of fully correct answers compared with the baseline, but it had the lowest number of fully supported answers. In other words, getting a better answer was not always the same as producing an answer that was clearly supported by the retrieved evidence. Evidence-constrained generation gave the strongest overall balance in this comparison.

5.4 Hallucination Analysis

I separately examined the hallucination annotations for the baseline and the two generation methods. The baseline annotation file contained two stale records that were excluded. Of the remaining 110 records, 108 had assessable labels and two I was uncertain. Among the 108 assessable baseline cases, 76 were labelled as containing hallucination, giving a rate of 70.37%.

For the method comparison, the hallucination rates were 70.37% for the baseline, 41.07% for evidence-constrained generation, and 58.04% for query rewriting. These figures suggest that evidence-constrained generation was associated with fewer hallucination labels in this experiment. The paired comparison between the baseline and evidence-constrained generation was statistically significant ($p < 0.001$). The baseline-to-query-rewriting comparison was not significant at the 0.05 level ($p = 0.0533$). Evidence-constrained generation also had a lower hallucination rate than query rewriting, with a significant paired difference ($p = 0.0043$).

Method	Labelled cases	Rate
Baseline	76/108	70.37%
Evidence-constrained	46/112	41.07%
Query rewriting	65/112	58.04%

Table 6: Hallucination rates across generation methods. The baseline rate uses 108 assessable cases after uncertain and stale records were excluded.

5.5 Effect of Reranking on Retrieval

The last part of the experiment was to see whether a cross-encoder reranker could improve the quality of the retrieved evidence. The system would first retrieve 20 candidate passages, then it used the cross-encoder to score the query-passage pairs. The five highest-scoring passages were selected for the final context.

Reranking increased mean Recall@5 from 66.52% to 77.68%, an improvement of 11.16%. The number of questions for which all supporting titles were retrieved increased from 36.61% to 58.93%. MRR@5 also increased from 0.8720 to 0.9092.

Metric	Before	After
Recall@5	66.52%	77.68%
All titles	36.61%	58.93%
MRR@5	0.8720	0.9092

Table 7: Retrieval performance before and after cross-encoder reranking. All titles refers to the proportion of questions for which every supporting title was retrieved.

The results indicate that reranking improved the retrieval metrics on this question set. The top-20 candidate pool had a mean Recall@20 of 80.80%, with all supporting titles retrieved for 65.18% of questions. This shows that some relevant passages were present in the initial candidate pool but did not make it into the top five until the reranking stage.

5.6 Summary of Results

Across the experiments, I found that changing question wording affected the retrieved passages, although the paired tests did not establish a statistically significant drop in complete evidence retrieval or answer correctness. Evidence-constrained generation gave the strongest answer-quality results among the generation methods. Query rewriting improved fully correct answers over the baseline, but its answers were less often fully supported by the retrieved evidence. Reranking improved the retrieval metrics, particularly the proportion of questions for which all supporting titles were retrieved.

Overall, the results tell us that retrieval quality and answer quality need to be evaluated separately. A change that improves one part of the pipeline might compromise on another.

6 Discussion

The results tell us that small changes in question wording can affect an RAG system at the retrieval stage, but the effect on the final answer is less significant. Among all 28 questions, the rewritten prompts often returned a different set of passages from the original questions. However, the differences in complete supporting-title retrieval were not statistically significant. This means that the experiment provides evidence of retrieval variation, but not enough evidence to conclude that paraphrasing consistently reduces retrieval performance.

6.1 Prompt Wording and Retrieval Stability

The first research question asked whether different versions of the same question would retrieve the same supporting passages. The results show us that retrieval was not fully stable across prompt versions. The average top five overlap with the original question ranged between 77.86% and 86.43% for the rewritten prompts.

This is important because dense retrieval systems represent questions as vectors. Even if two questions have the same meaning, differences in framing the sentence could produce different vector representations. The retriever then ranks passages differently. The results from this experiment are consistent with this hypothesis.

However, changes in the retrieved passages do not always mean that useful evidence has been lost. Some passages might contain similar information or give enough context to answer the question asked, even if they are not the exact supporting passages identified in the dataset. This might explain why the retrieval overlap changed a lot more than the answer correctness scores.

6.2 Retrieval Changes and Answer Quality

The second research question looked at whether retrieval differences were visible in answer correctness and evidence support. The answers did not show a statistically significant difference between the original questions and their rewritten versions. The number of fully correct answers was around six to eight across the four prompt versions. Fully supported answers also varied, but the differences

were small, so there is little meaning comparing them.

One potential hypothesis is that the language model can sometimes produce the correct answer even when the retrieved passages differ from the expected evidence. It could find enough information in an alternative passage, or it may make use of knowledge that it learned during pre-training. This can make the final answer appear correct even when the retrieval results are not supporting it.

This also creates a problem when evaluating RAG systems. A correct answer is not necessarily a grounded or factually 100% accurate answer. If the model produces the right response without sufficient support in the retrieved context, the system is still unreliable. This is why I have separately evaluated fully correct and fully supported answers in the experiments.

The failure categories give us another view of this issue. Missing retrieved information and generation errors appeared at similar frequencies in the answer audit. This means that problems were not just limited to the retriever. Even when relevant information was available, the model sometimes failed to use it correctly or produced an incomplete response. Improving retrieval alone is thus not enough to improve the complete RAG pipeline.

6.3 Evidence-Constrained Generation and Query Rewriting

The third question checked if evidence-constrained generation and query rewriting could improve answer quality and reliability. Evidence-constrained generation produced the strongest overall results among all the methods. It produced more fully correct and fully supported answers than the baseline with a lower hallucination rate.

This means that giving the model clearer or more straightforward instructions about how to use the retrieved passages can help. Asking the model to stay within the evidence it can retrieve and acknowledge when the context is insufficient also reduces unsupported responses. It does not remove every error, but it does improve overall answer reliability and its relation with the retrieved documents.

Query rewriting gave a somewhat mixed result. It increased the number of fully correct answers compared with the baseline, but it produced fewer fully supported answers. This might be because rewriting can make a query more useful for retrieval, but it might also change which details receive emphasis. If the rewritten query retrieves passages that are related to the question but do not contain all the required evidence, the model probably will generate an answer that is not fully supported.

This difference is important because query rewriting is often looked at as a general or traditional way to improve retrieval. The results here mean that this "improvement" should not be evaluated only by whether it produces a correct final answer. The evidence used to produce that answer also matters. In my experiment, evidence-constrained generation gave a better balance between correctness and support.

6.4 The Effect of Reranking

The fourth point to be examined is whether reranking could improve the selection of supporting passages. The reranking experiment showed improvements across all three retrieval measures. In particular, the proportion of questions for which all supporting titles were retrieved increased after reranking. Thats why I decided to make reranking its own subsection.

The top 20 candidate results actually do explain this improvement. A larger candidate pool had more of the required evidence than the initial top five results. The cross-encoder could then reorder these candidates so that more relevant passages would appear in the final context. This means that some retrieval failures were not caused by the lack of useful passages from the candidate pool, but instead by their initial ranking.

Thus, reranking seems useful for improving the evidence supplied to the generator. However, better retrieval does not always mean a better answer. The generator still needs to identify the relevant information and use it correctly.

6.5 Overall Interpretation

Essentially, the experiments show that RAG performance depends on several connected stages. Prompt wording can change the retrieved passages. Retrieval quality can affect what evidence is available to the generator. The generation instructions then influence how the model uses that evidence.

The results also prove that a single evaluation measure is simply not enough. Retrieval metrics describe whether useful passages were found. Answer correctness measures whether the response matches the expected answer. Evidential support measures whether the response can be justified

from the supplied context. Each of these metrics is important in their own way. Improvement in any one of these metrics might be a trade off, losing at another metric.

For this particular dataset, evidence-constrained generation had the strongest answer-quality results, while reranking had the strongest retrieval improvements. Neither result gave us a universal solution, or a final "better" method.

7 Conclusion

In this project, I looked at whether changing the wording of a question affects the passages retrieved by a RAG system, and whether those changes also affect the answers it produces. I used 28 questions from HotpotQA, with three rewritten versions of each question, and compared the system's retrieval and answer quality across the different prompts.

The main finding was that changing the wording did actually affect retrieval. The rewritten questions often strayed from the same top five passages as the original questions. In spite of this, there was no statistically significant difference in complete evidence retrieval or answer correctness between the prompt versions. Thus, even though the retriever was affected by wording, the effect on the final answers was very less.

I also tested a few ways to improve the system. Evidence-constrained generation was the best one for answer correctness and evidence support. Query rewriting improved fully correct answers, but the answers were rarely fully supported by the retrieved passages. Reranking improved the retrieval scores, as well as increased the number of questions for which all supporting titles appeared in the top five.

One important point is that an RAG system cannot be judged only by whether its final answer is correct. It also matters whether the system retrieved the right information. These are separate parts of the process, and an improvement in one does not always mean an improvement in the others. It is mostly a tradeoff.

The project was limited by its small question set, the model used, and the manual answer evaluation. The results would probably be different, but following a similar pattern on a larger set of questions and with other language models.

Limitations

There are a few limitations to keep in mind when looking at these results. The biggest one is the size of the final evaluation set. Although I started with a sample of 1,000 HotpotQA questions, the final experiments used only 28 questions. I removed two questions after checking whether their supporting passages actually contained enough information to establish the expected answer. This left 17 bridge questions and 11 comparison questions.

I did this because I did not want to evaluate the system on questions where the evidence itself was unclear. However, this also means that the final set is small and curated. It does not represent the full range of questions in HotpotQA. Simply put, this experiment is more representative of a batch of hotspotQA questions, not the set in its entirety.

Another limitation is the language model used for answer generation. I used Qwen2.5-0.5B-Instruct, which is a relatively small model. My GPU VRAM was not sufficient to run higher models, and running it on my CPU would take hours per step per stage of the experiment. The reduced size made the experiments manageable, but it probably also have affected the answers. Some failures may have been caused by the model's ability to understand and combine information, rather than by retrieval alone. A larger model might produce different results.

The passage collection is also a discussion point. I extracted the passages in the context fields of the 1,000 sampled HotpotQA examples, combined them into a single collection, and removed duplicate ones. This left me with 9,758 unique passages. But the retrieval results may be different when the system has to search through a much larger collection containing more irrelevant passages or more competing or relevant passages.

There are also limits to the way I checked and reviewed the answers. I reviewed the answers myself, but my assessments are subjective in nature, even though I did my best to compare evidence as objectively as possible. Some answers were also difficult to judge because they were either incomplete or they simply did not show whether a claim was supported. I excluded these uncertain and stale records from the baseline hallucination calculation entirely. That is why I have judged a total of 108 answers rather than 112.

A larger scale on this project could fix several of these issues by using more questions, testing ad-

ditional language models, and having independent annotators or third party reviews for the answers. It would also be useful to test the same methods on a broader passage collection.

Ethics Statement

This project used the HotpotQA dataset for research and evaluation of a retrieval-augmented question answering system. The generated answers were checked to evaluate their correctness and support from the retrieved passages. I only used manual reviews for the evaluation, and sometimes when there were ambiguous cases, I used AI to proofread my evaluation, albeit rarely, since AI is not reliable for a project like this. Since the evaluation labels were subjective, even though I did my best to be objective, they should not be treated as absolute objective or 100% correct judgements, since human error might exist. This entire project was conducted for my course DA624: NLP in LLMs, and the results are reported taking into account the limitations of the dataset and evaluation process.

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A Question Variants Used in the Worked Example

The example in this appendix is from the final set of 28 HotpotQA questions. The original question and its three rewritten versions are shown below.

Original question:

The lead singer and guitarist in the Portland, Oregon rock band consisting of John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist and Eric Howk, goes by what alias?

Paraphrase 1:

What alias is used by the lead singer and guitarist of the Portland, Oregon rock band whose members include John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk?

Paraphrase 2:

What is the alias of the frontman who sings and plays guitar in the Portland rock band made up of John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk?

Paraphrase 3:

The Portland, Oregon band comprising John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk has a lead singer and guitarist known by what alias?

B Additional Experimental Settings

The retrieval system used the embedding model `sentence-transformers/all-MiniLM-L6-v2`. Passage embeddings were normalized before being indexed using **FAISS** `IndexFlatIP`. The retriever returned the top five passages for each question.

For answer generation, I used `Qwen/Qwen2.5-0.5B-Instruct` with the Hugging Face Transformers library. Generation used **greedy**

decoding (`do_sample=False`), a maximum of 48 new tokens, and a repetition penalty of 1.05.

The reranking experiment used `cross-encoder/ms-marco-MiniLM-L-6-v2`. It reranked the initial set of 20 retrieved candidates and selected the top five passages for the final context.

C Worked Example of Prompt Variability

This example shows how changing the wording of a question affected the retrieved passages and the generated answer. It compares the original question with Paraphrase 3 from the previous section.

C.1 Original Question

The original question asked for the alias of the lead singer and guitarist of the Portland, Oregon rock band consisting of John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk.

The expected answer was *The Fantastic The*. The two supporting titles listed for this question were *John Gourley* and *Portugal. The Man*.

The top five retrieved titles were:

1. John Gourley
2. Scott Devours
3. Enemymine
4. Mocean 8
5. Portugal. The Man

Both supporting titles appeared in the retrieved results. The generated answer was:

> The lead singer and guitarist in the Portland, Oregon rock band consisting of John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist and Eric Howk, goes by the alias "The Fantastic The."

This answer gave the expected alias and was marked correct and supported in the answer review.

C.2 Rewritten Question

Now we compare the original question with Paraphrase 3:

> The Portland, Oregon band comprising John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk has a lead singer and guitarist known by what alias?

The top five retrieved titles changed to:

1. Scott Devours
2. Enemymine
3. Joey Jordison
4. John Gourley
5. Brian Robertson (trombonist)

The passage titled *Portugal. The Man* was no longer retrieved. As a result, only one of the two supporting titles was present in the top five results. Recall@5 decreased from 1.0 for the original question to 0.5 for this rewritten version.

The generated answer was:

> The lead singer and guitarist of the Portland, Oregon band comprising John Gourley, Zach Carothers, Kyle O'Quin, Jason Sechrist, and Eric Howk is John Baldwin Gourley.

This response identified the person but did not provide the alias requested in the question. The answer review marked it as partly correct, supported, and incomplete.

C.3 What This Example Shows

The two versions returned different sets of passages. The original question retrieved both supporting titles, whereas Paraphrase 3 retrieved only *John Gourley*. This changed the retrieval, even though the relevant information was still available.

In fact, the retrieved biography of John Gourley contained the information needed to answer the question. It described him as the lead singer and guitarist of **Portugal. The Man** and stated that he also uses the alias **The Fantastic The**. The rewritten version thus did not give an incorrect answer, just a partly correct answer.

However, the generated answer gave his full name instead of the alias. This shows us that retrieval and answer generation can fail in different ways. A change in retrieved passages does not make the final answer correct. The model also needs to identify the specific information requested and return it accurately.